1.7 Chaotic Trajectories of Nonlinear Systems

Restricting our study to the curves defined by a system of first integrals and/or the consideration of small oscillations near equilibrium points would greatly limit the range of accessible dynamical phenomena. If we construct a molecular system describing many bodies interacting in nonlinear force laws, there would be no obvious reason to expect that there are many first integrals present to constrain the motion. Indeed, experience indicates that the situation is quite the opposite: typically only the energy and the total momentum of a molecular system are conserved quantities. Thus we are in general left with a complicated dynamical system whose solutions cannot be analytically determined.

Example 1.8[5] *(Planar Lennard-Jones Trimer)* One of the simplest illustrations of the chaotic nature of molecular systems is given by the Lennard-Jones model consisting of just three atoms with motion restricted to the plane. The energy is

$$E = K + U = \frac{\|\dot{\boldsymbol{q}}_1\|^2}{2} + \frac{\|\dot{\boldsymbol{q}}_2\|^2}{2} + \frac{\|\dot{\boldsymbol{q}}_3\|^2}{2} + \hat{\varphi}_{\mathrm{LJ}}(\|\boldsymbol{q}_1 - \boldsymbol{q}_2\|) + \hat{\varphi}_{\mathrm{LJ}}(\|\boldsymbol{q}_2 - \boldsymbol{q}_3\|) + \hat{\varphi}_{\mathrm{LJ}}(\|\boldsymbol{q}_1 - \boldsymbol{q}_3\|),$$

with the interatomic interaction given by $\hat{\varphi}_{\mathrm{LJ}}(r) = 4[r^{-12} - r^{-6}]$.

The global minimum of this simple system must be radially symmetric. Placing the atoms at the vertices of an equilateral triangle, we have

$$U = 3\hat{\varphi}_{\mathrm{LJ}}(r),$$

where r is the length of a side. This is minimized when $r = 2^{1/6} \approx 1.1225$.

In general, when the total potential is a sum of distance potentials

$$U = \frac{1}{2}\sum_{i \neq j} U_{ij},$$

where $U_{ij}(\boldsymbol{q}_i, \boldsymbol{q}_j) = \varphi_{ij}(\|\boldsymbol{q}_i - \boldsymbol{q}_j\|)$, we say that the system has *central forces*. In this case,

$$\frac{\partial}{\partial \boldsymbol{q}_i} U_{ij}(\boldsymbol{q}_i, \boldsymbol{q}_j) = -\frac{\partial}{\partial \boldsymbol{q}_j} U_{ij}(\boldsymbol{q}_i, \boldsymbol{q}_j),$$

[5] The Lennard-Jones Trimer is included as an example in the **MD.M** package, see http://MolecularDynamics.info